

Astrometry Feature Additions in `gulls_mp`

This document records the astrometric light-curve machinery that was introduced after branching from `dev`. The focus is on the mathematical model, reference-frame bookkeeping, and observable products, with explicit call-outs for places where the current implementation looks fragile or incorrect.

1. Scope and Control Surface

- New runtime switches are parsed from the parameter file (`ASTROMETRY_ON` , `ASTROMETRIC_SYS_FLOOR` ; `src/readParamfile.cpp:234`) and cached on the simulation state (`struct filekeywords` , `src/structures.h:140`).
- Event buffers now carry sky-frame centroid tracks and uncertainties (`cNtrue` , `cEtrue` , etc.; `src/structures.h:386 – 392`), which are resized alongside the legacy photometry arrays (`src/timeSequencer.cpp:231 – 257`).
- The main generator toggles VBMicrolensing’s internal centroid solver through `Event->vbm->astrometry` (`src/pllxLightcurveGenerator.cpp:57`) and provides the data needed to rotate lens-frame centroids into the observable North/East frame.
- Downstream outputs and plotting utilities consume these new columns:
 - `outputLightcurve.cpp` appends NE centroids and absolute RA/Dec tracks to the `.lc` products (`src/outputLightcurve.cpp:380 – 436`).
 - `smoke_test/plotting.py` overlays VBMicrolensing predictions and draws diagnostic vectors for the proper-motion budget (`smoke_test/plotting.py:208 – 269` , `smoke_test/plotting.py:536 – 638`).

2. Lens-Frame Modelling

Let the intrinsic event parameters be the usual binary-lens tuple $(s, q, u_0, \alpha, \rho, t_E, t_0)$ augmented by the Einstein radius θ_E and the source/relative proper-motion metadata already present in GULLS.

2.1 Source trajectory in the lens plane

For an epoch t_i the scaled time coordinate is

$$\tau_i = \frac{t_i - t_0}{t_E^{(\text{ref})}} + \delta\tau_i,$$

where the parallax correction

$$\delta\tau_i = \Delta t_{\text{par},i}$$

is injected when `PARALLAX` is enabled (`src/pllxLightcurveGenerator.cpp:116 – 123`). The impact-parameter component is

$$u_i = u_0 + \delta u_{\text{par},i}.$$

Coordinates in the binary lens frame (with $\hat{x}_1$ along the lens axis and $\hat{x}_2$ perpendicular) follow the standard rotation by the trajectory angle α :

$$\begin{aligned} x_{s,i} &= \tau_i \cos \alpha - u_i \sin \alpha + x_{\text{CoM}}, \\ y_{s,i} &= \tau_i \sin \alpha + u_i \cos \alpha, \end{aligned}$$

where the barycentric shift of the primary lens is $x_{\text{CoM}} = -(1 - m_1)s$ (`src/pllxLightcurveGenerator.cpp:117 – 134`).

2.2 VBMicrolensing interface

With `Event->vbm->astrometry` flagged, `BinaryMag2` returns both magnification and the instantaneous centroid in Einstein radii:

$$(\xi_1, \xi_2) = (\text{Event->vbm->astrox1}, \text{Event->vbm->astrox2}).$$

For single sources the lens-frame centroid is simply ξ (`src/pllxLightcurveGenerator.cpp:200 – 209`). For a binary source the code re-evaluates VBMicrolensing for the companion trajectory and forms the flux-weighted centroid

$$\xi_{\text{tot}} = \frac{F_1 \xi_1 + F_2 \xi_2}{F_1 + F_2},$$

with $F_1 = A_1$ and $F_2 = f_{s,2/1} A_2$ (`src/pllxLightcurveGenerator.cpp:162 – 176`). In all cases the lens-frame centroid is stored (for smoke tests) as `xctrue` , `yctrue` .

3. Rotation into the North/East frame

The Einstein-radius scaling converts to milliarcseconds:

$$(c_1, c_2) = \theta_E (\xi_1, \xi_2).$$

To project into geocentric North/East coordinates the code solves for a rotation angle Φ between the lens axis and North:

1. Prefer the direction of π_E :

$$\Phi = \text{atan2}(\pi_{E,E}, \pi_{E,N}) - \alpha.$$

This path is taken when `PARALLAX` is active (`src/pllxLightcurveGenerator.cpp:62 – 69`).

2. Otherwise fallback to the relative proper-motion vector. The Galactic components (μ_ℓ, μ_b) are converted to Equatorial $(\mu_{\alpha^*}, \mu_\delta)$ via `coords.mulb2ad` (`src/pllxLightcurveGenerator.cpp:71 – 78`). Φ is then set to $\text{atan2}(\mu_{\alpha^*}, \mu_\delta) - \alpha$.

If neither vector is available the rotation is left at the identity and astrometric products are suppressed (`astrom_ok = false`).

Using Φ , the code applies the right-handed rotation matrix

$$R(\Phi) = \begin{pmatrix} \cos \Phi & \sin \Phi \\ -\sin \Phi & \cos \Phi \end{pmatrix},$$

so that

$$\begin{pmatrix} \Delta N \\ \Delta E \end{pmatrix} = R(\Phi) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

The results are cached in `cNtrue`, `cEtrue` (`src/pllxLightcurveGenerator.cpp:221 – 226`), while `cNobs`, `cEobs` start as noiseless copies.

4. Absolute sky coordinates and light-curve output

Prior to writing the `.lc` file the code adds the lens heliocentric proper motion to the microlensing offsets and converts into RA/Dec:

$$\begin{aligned}\Delta N_{\text{tot}}(t_i) &= \Delta N_{\text{micro}}(t_i) + \mu_\delta \Delta t_i, \\ \Delta E_{\text{tot}}(t_i) &= \Delta E_{\text{micro}}(t_i) + \mu_{\alpha^*} \Delta t_i,\end{aligned}$$

with $\Delta t_i = (t_i - t_0)/365.25 \text{ yr}$ (`src/outputLightcurve.cpp:380 – 386`).

Assuming small angles,

$$\begin{aligned}\alpha_{\text{true}}(t_i) &= \alpha_0 + \frac{\Delta E_{\text{tot}}(t_i)}{\cos \delta_0} \frac{1}{3600 \cdot 10^3}, \\ \delta_{\text{true}}(t_i) &= \delta_0 + \frac{\Delta N_{\text{tot}}(t_i)}{3600 \cdot 10^3},\end{aligned}$$

where α_0, δ_0 are the event coordinates from the Galactic-to-equatorial transform (`src/timeSequencer.cpp:20 – 33`). Observed quantities reuse the same transformation with `cEobs` , `cNobs` , and the positional errors reuse the same scaling (`src/outputLightcurve.cpp:394 – 401`). All of these arrays are streamed to the light-curve file with explicit column names (`src/outputLightcurve.cpp:409 – 436`).

5. Noise and systematics model

After photometry, the astrometric measurement is perturbed by an axisymmetric Gaussian whose width combines photon statistics with a configurable floor:

$$\sigma_{\text{axis}} = \sqrt{\left[\frac{\text{FWHM}}{\sqrt{8 \ln 2}} \sigma_{\text{phot}} \right]^2 + \sigma_{\text{sys}}^2},$$

where $\sigma_{\text{phot}} = |\sigma_F/F|$ is derived from the measured flux errors and σ_{sys} is

`ASTROMETRIC_SYS_FLOOR` (`src/photometry.cpp:134 – 152`). The observed centroids are then drawn as

$$c_{N,\text{obs}} = c_{N,\text{true}} + \sigma_{\text{axis}} \mathcal{N}(0, 1), \quad c_{E,\text{obs}} = c_{E,\text{true}} + \sigma_{\text{axis}} \mathcal{N}(0, 1),$$

with matching 1- σ errors reported (`src/photometry.cpp:154 – 158`). In “ideal” photometry mode the code bypasses the random deviate and simply forwards the truth values while pinning the error budget to the systematic floor (`src/photometry.cpp:102 – 121`).

Open bug: the noise draw is skipped whenever $c_{N,\text{true}} = c_{E,\text{true}} = 0$. Exact symmetry points (e.g. at large $|t|$ or during axis crossings) therefore record zero observed shift with zero uncertainty, even though centroiding noise should still be present (`src/photometry.cpp:154 – 164`). This yields discontinuities in the time series and biases the RA/Dec errors.

6. Smoke-test instrumentation

The smoke test ingests the new columns to produce three coordinated diagnostics per light curve:

1. **Lens-frame overlay.** `BinaryAstroLightCurve` from `VBMicrolensing` is evaluated a second time with the summary metadata and compared against the stored lens-frame truth (`smoke_test/plotting.py:208 – 269`). Because the frame mapping is uncertain, the script brute-forces all eight permutations/sign flips of the VBM output in search of the smallest MSE difference from the simulated centroid (`smoke_test/plotting.py:233 – 256`). This hack is a strong indicator that the rotation logic above is still inconsistent with the `VBMicrolensing` convention.
2. **North/East panel.** `true_N/E` and `measured_N/E` are plotted alongside the VBM sky-track, using the new error bars (`smoke_test/plotting.py:559 – 614`).
3. **Absolute RA/Dec panel.** The code reconstructs the absolute sky track and draws diagnostic vectors for the geocentric relative motion, plus the heliocentric source and lens proper motions when available (`smoke_test/plotting.py:536 – 556` , `smoke_test/plotting.py:620 – 638`). Direction arrows span the temporal extent of the light curve to make frame mix-ups easier to spot.

Any failure to build the VBM overlay when astrometry is expected triggers the smoke test (`smoke_test/plotting.py:870 – 909`).

7. Additional gaps and risks

1. **Frame alignment remains ambiguous.**

The need for the sign/permutation search in the smoke test (`smoke_test/plotting.py:233 – 256`) implies that the rotation angle Φ may still be defined in the wrong sense or with a missing reflection. Comparing against the BAGLE frame-conversion notebooks—particularly `BAGLE_Microlensing/docs/frame_convert.rst`—would help pin down whether $\hat{x}_1$ should map to North or East after subtracting α .

2. **Zeroing out astrometry when orientation is unknown.**

If neither π_E nor μ_{rel} is available, `astrom_ok` stays false and all centroids are forced to zero (e.g. events without parallax or catalog proper motions;

`src/pllxLightcurveGenerator.cpp:80 – 89`). The `.lc` file still carries the lens proper-motion drift, so the RA/Dec columns evolve while the NE offsets are identically zero: this mixture is confusing for downstream inference.

3. Multiple-source weighting ignores baseline mixing.

The flux-weighted centroid uses $F_1 = A_1$ and $F_2 = f_{s,2/1} A_2$ (`src/pllxLightcurveGenerator.cpp:168 – 172`). The total flux in the denominator is $A_1 + f_{s,2/1} A_2$, whereas the photometric magnification stored in `At rue` subtracts the baseline contribution for source 2 (`amp + f_s (A_2-1)`). This mismatch can bias the centroid when the secondary contributes significant baseline light.

4. Proper-motion conversion assumptions.

The conversion from Galactic to Equatorial proper motion relies on `coords.mu1b2ad` returning μ_α^* (i.e. already multiplied by $\cos \delta$; `src/classes/coords.cpp:272 – 294`). If the upstream catalog stores μ_α without the cosine factor the RA drift will be off by $\cos \delta$. Cross-checking against the Astrometry repo’s derivations (`Astrometry/docs/astrometry.tex`) is recommended.

5. RA/Dec suppression when astrometry off.

When `ASTROMETRY_ON=0` , the `.lc` columns for RA/Dec default to zero (`src/outputLightcurve.cpp:388 – 407`). Downstream tooling that expects absolute coordinates will therefore see the event at the origin unless those columns are explicitly ignored.

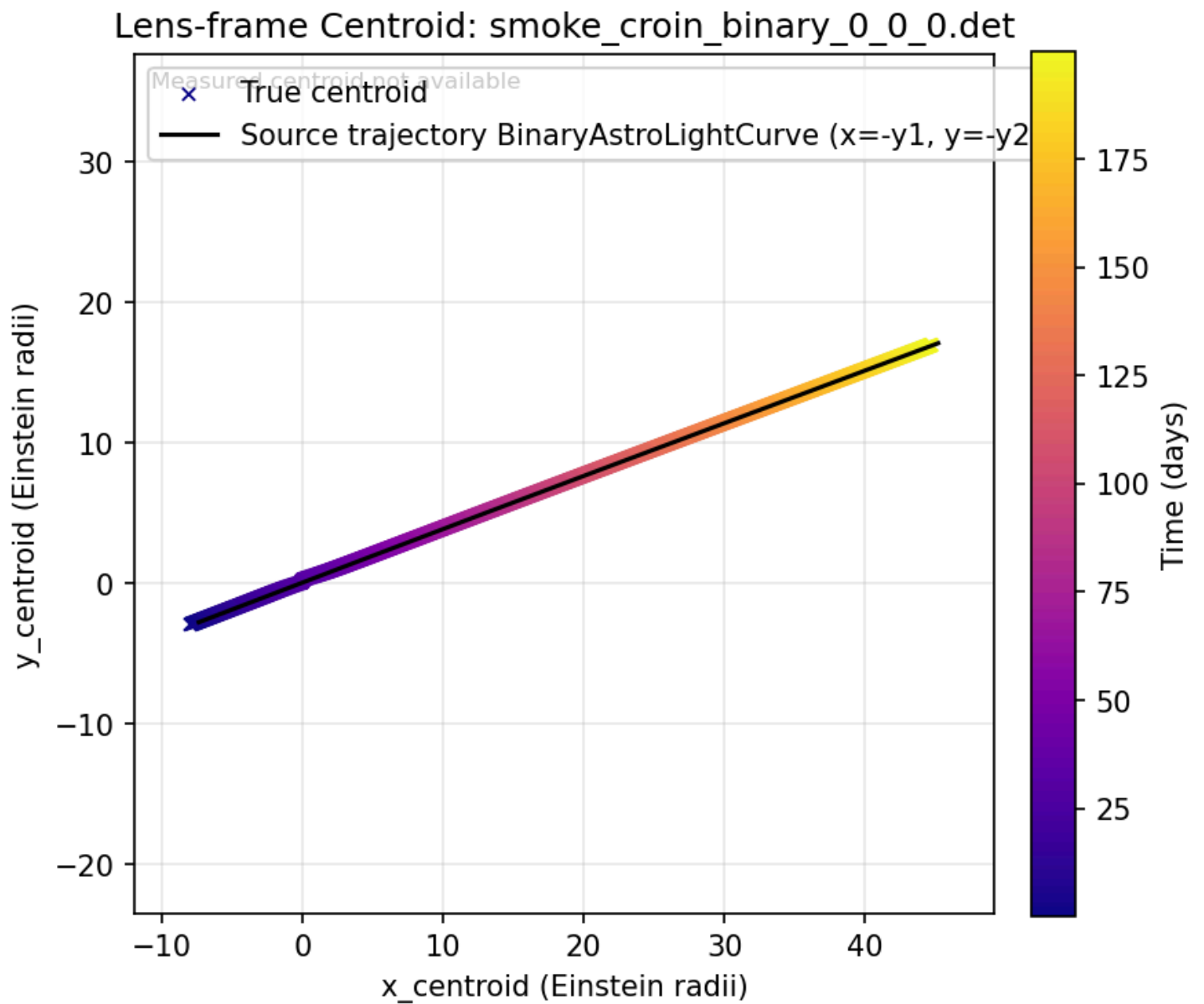
8. Suggested validation steps

1. Use the BAGLE `Frame_conversion_examples.ipynb` to derive the expected NE centroids for a known configuration and compare against GULLS output with identical parameters.
2. Modify the smoke test to log the best permutation it selects; this will expose the systematic sign needed to fix the rotation.
3. Inject a synthetic event where $\Delta N = \Delta E = 0$ for all times but request astrometry to verify that the noise model produces finite uncertainties; this will confirm the bug in Section 5 before patching.
4. Cross-check the RA/Dec drift against the Astrometry repo’s analytic model (equations in `Astrometry/docs/astrometry.tex`) to ensure the lens proper motion is not double-counted.

References

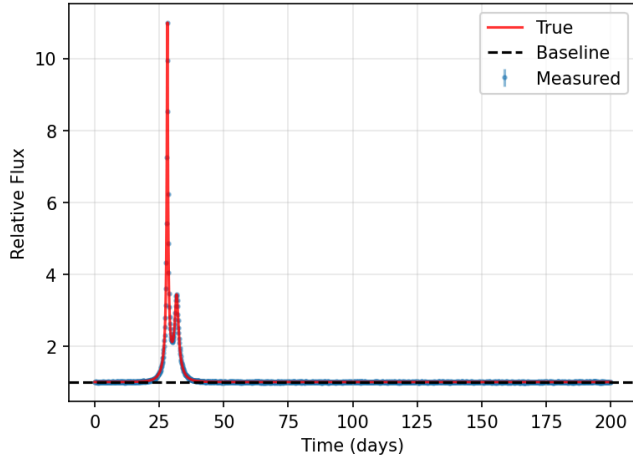
- VBMicrolensing centroid solver: `src/classes/VBMicrolensingLibrary.cpp` (numerous `astrox{1,2}` code paths).

- Error-model derivation followed from Astrometry collaboration notes (Astrometry/docs/astrometry.tex).
- Frame-convention comparisons are available in BAGLE_Microlensing/docs/frame_convert.rst .

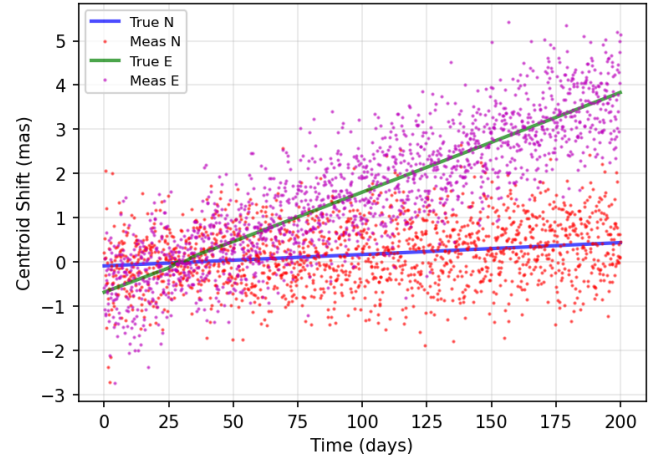


Smoke Test: smoke_croin_binary_0_0_0.det
 Lens M=0.103 Msun, Lens D=4.883 pc, Source D=5.076 pc, $\theta_E=0.081$, $\mu_{\text{rel}}=(-5.376, 6.291)$ mas/yr

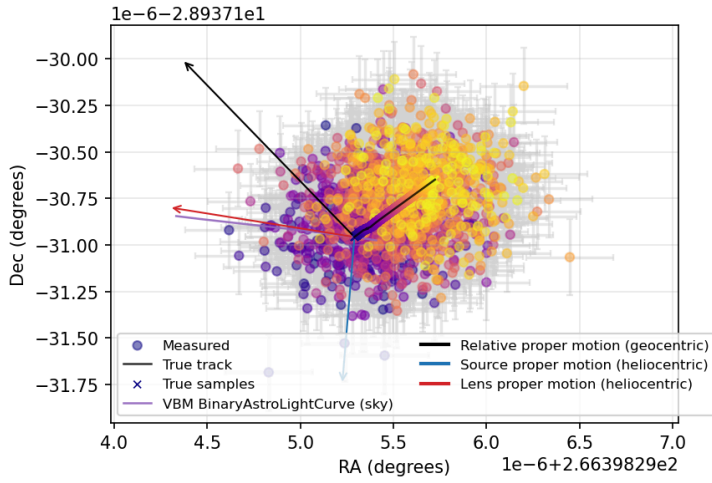
Light Curve



Astrometric Timeseries (N and E)



Absolute Astrometric Position



Astrometric Centroid (N/E), Relative to the Lens

